b)

GS mep: $$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}^{(k+1)} = \begin{bmatrix} 0 & 0 & 0 \\ -1/3 & 0 & 0 \\ -2/5 & -2/5 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}^{(k+1)} + \begin{bmatrix} 0 & -1/5 & -2/5 \\ 0 & 0 & 1/3 \\ 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}^{(k)} + \begin{bmatrix} 8/5 \\ 1 \\ 9/5 \end{bmatrix}$$ ✓

$x^{(0)} = (1.01, 0.99, 1)^T$

$$x_1^{(k+1)} = (-1/5)x_2^{(k)} + (-2/5)x_3^{(k)} + 8/5$$ ✓

$$x_2^{(k+1)} = (-1/3)x_1^{(k+1)} + (1/3)x_3^{(k)} + 1$$ ✓

$$x_3^{(k+1)} = (-2/5)x_1^{(k+1)} + (-2/5)x_2^{(k+1)} + 9/5$$ ✓

k	x_1	x_2	x_3
0	1.01	0.99	1
1	1.0020	0.99933	0.99946

✓

4p†

$$x_1^{(1)} = (-0.2)(0.99) + (-0.4)(1) + 1.6$$
$$= 1.002$$ ✓

$$x_2^{(1)} = (-0.33333)(1.002) + (0.33333)(1) + 1$$
$$= 0.999\,33$$ ✓

$$x_3^{(1)} = (-0.4)(1.002) + (-0.4)(0.99933) + 1.8$$
$$= 0.999\,46$$ ✓

$$x^{(1)} = \begin{bmatrix} 1.002 \\ .99933 \\ 0.99946 \end{bmatrix}$$ ✓